

The Routing Campaign

Wave 4 of the tempotron foundations program: what a stuck learner is doing,
what the comb-kernel lift is, and what constraining the temporal baseline reveals
(2026-08-23)

snn-research / tempotron-capacity program

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Abstract

Waves 1–3 of this program (companion report, 2026-08-21) established that the tempotron’s findability wall is a search-time phenomenon in a solution space whose cells are certified plentiful, and that gradient descent is a late-settling, margin-biased router. Wave 4 (“ROUTING”, three verifier-gated workstreams, $\approx$ \$16 lab spend) asked three questions and answered all three. (1) *What is a non-converging learner doing?* It is a localized glassy wanderer — no limit cycle, no frozen frustrated core — that repeatedly *stands inside* rooms LP-certified to contain exact solutions (1.97 million visits across stuck runs) and is ejected after a median of one epoch, predominantly by minus-pattern updates: the obstruction is the learning rule’s within-room dynamics, not existence, reachability, or exploration. Convergence is near-deterministically decided by the drawn instance (14/16 instances all-or-nothing, $p \sim 10^{-13}$). (2) *What is the comb-kernel capacity lift?* An interior maximum of findable capacity in the effective rank of the field covariance (peak at ≈ 3 –4 spectral lines), collapsing exactly where lines become unresolvable in the coding window — predicted pre-data by a finite-rank theory — and carried by a channel provably invisible to replica-symmetric geometry: decision-time *multiplicity*. (3) *What do constraints on a learnable temporal baseline do?* At the capacity level, almost nothing — one principle (only the uniform mode has first-order value, and tonic *inhibition* owns it) renders every studied constraint a null — but at the learner level they are live regularizers: a theory-predicted *sign reversal* of the effect of frozen offsets was confirmed ($p = 1.5 \times 10^{-5}$), and balance-constrained learned shapes produce gains where both capacity criteria price shape at exactly zero. Completing the criterion adjudications shows that *no existing analytic criterion describes the findable surface*; its anisotropy is now measured directly.

1 The stuck-model autopsy (WS-AUTOPSY)

Pre-registered alternatives for a non-converging learner: limit cycle, drifting random walk, or frustrated-core churn. All three were scored against per-epoch decision-time (“room”) trajectories at scale and against exact ground truth at toy scale, with four independent verifier gates.

Not a cycle, not a small core. Zero nontrivial cycles in 0/88 stuck toy twins (two detectors, exhaustive verifier rescan) and zero room-vector recurrences in 313k logged pairs at scale; the wander is localized (median revisit 0.82) yet vast — 7.8k to 38.5k distinct rooms per stuck run, still growing at 64k epochs (Figs. 1, 2). The frustrated-core hypothesis fired its falsifier 88/88: violations sweep essentially all patterns (median 95.6% within even 100-epoch windows) — global churn, not a stuck kernel of hard patterns (Fig. 3).

The headline: ejection from certified-solvable rooms. 90.9% of stuck twins repeatedly stand *inside* rooms that an LP certifies to contain exact solutions — 1.97 million such visits —

and exit after a median of **one** epoch, with 55.9% of the exiting updates caused by minus patterns. Certificates were independently re-proven (84/84 exact LP re-solves). **The findability gap is caused by the rule’s within-room dynamics**: the learner finds the right rooms constantly; the update rule throws it out before the (convex) within-room problem can settle (Fig. 4).

Twins. Convergence is near-deterministically instance-determined: 14/16 instances had all eight seeds converge or all eight fail ($p = 2.1 \times 10^{-13}$). The apparent early-warning signal found in the free pass (AUC 0.897) resolved into an instance-hardness readout — within-instance prediction is null ($p = 0.54$) (Fig. 5).

Algorithmic corollary (registered for Wave 5, untested). A feasibility check or within-room convex polish before permitting exit updates should convert a large fraction of stuck runs. Spend: \$0.67 of \$6.

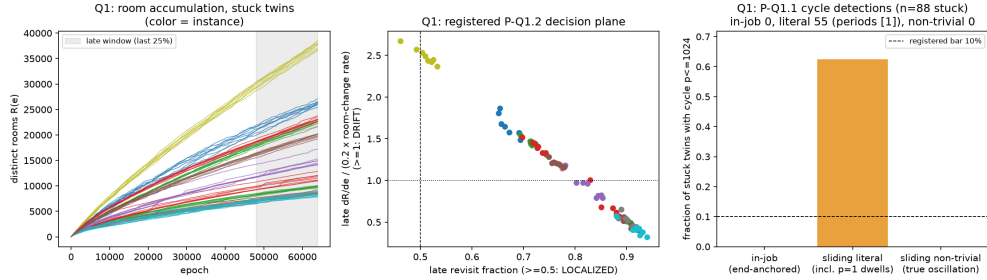


Figure 1: Toy-scale room-visitation statistics for stuck twins: localized but vast wander; no cycles.

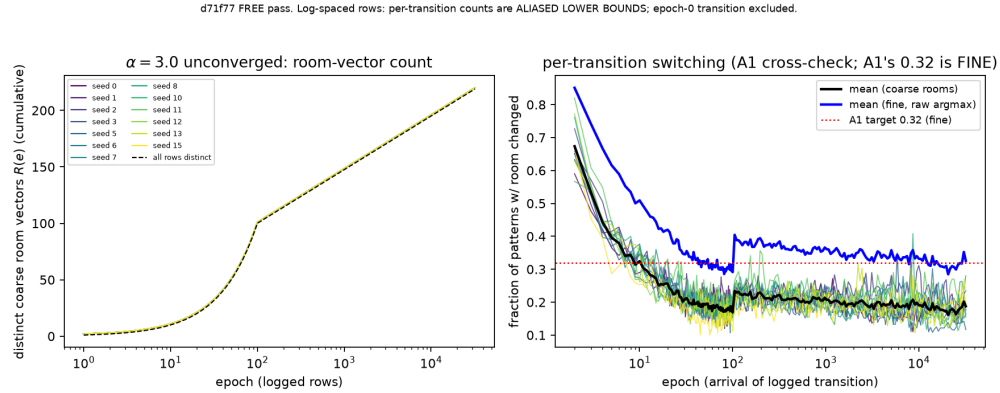


Figure 2: At-scale room dynamics: no recurrence over 313k logged pairs; switching never freezes for stuck seeds.

2 The comb-kernel lift is a rank resonance (WS-COMB)

Wave 3 found both comb kernels $\approx +0.45$ above the smooth-kernel anchor at matched effective dimension, unexplained by any registered theory. Wave 4 first discharged the mandatory instrumentation audit: the v13 harness’s `seeds=` label fault bit only the (already-excluded) comb waves; the June kernel campaigns are clean; the sinc anchor was, however, a mislabeled $n=8$ measurement (corrected: 3.286 [3.240, 3.300]). The lift survives the audit, and the harness is fixed and gated.

A finite-rank theory (timestamped before data) predicted an interior maximum in the number of spectral lines m with collapse once lines become unresolvable in the coding window ($m^* =$

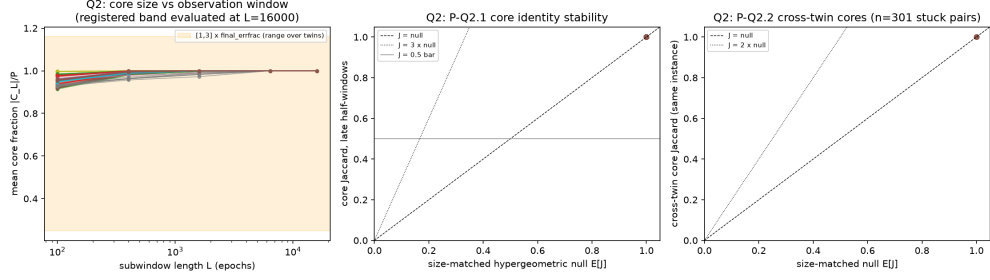


Figure 3: Frustrated-core test: violations sweep essentially all patterns — the registered falsifier fires; stuckness is global churn.

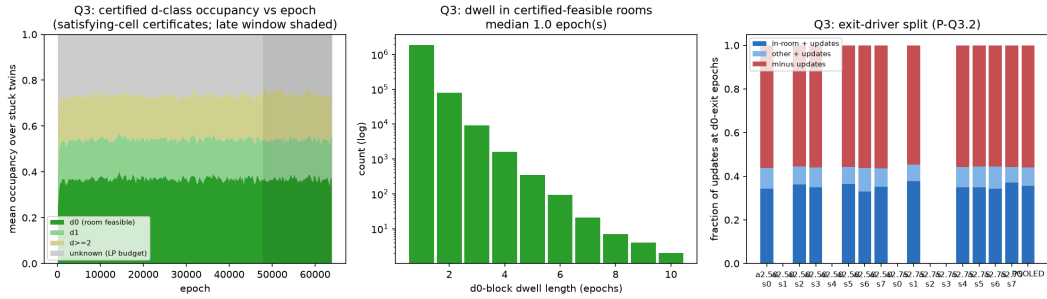


Figure 4: Near-miss analysis: stuck trajectories repeatedly stand inside LP-certified solution-containing rooms and are ejected within ~ 1 epoch — the rule-indicting outcome in its sharpest form.

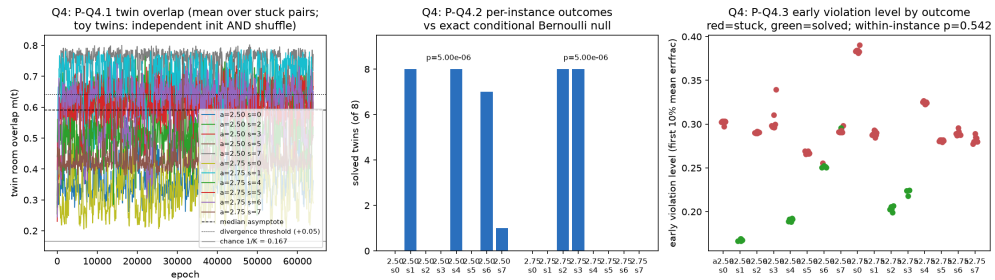


Figure 5: Twin studies: convergence is an instance property, not seed luck.

$\frac{1}{2}\sqrt{1 + 3K_{\text{eff}}^2/\pi^2} = 8.83$). The measured ladder ($n=16$ true seeds per cell) gives

$$\hat{\alpha}_c(m=1, 2, 4, 8, 16) = 3.522, 3.553, \mathbf{3.747}, 3.200, 3.240,$$

an interior maximum at effective covariance rank $D_{PR,\Sigma} \approx 10$ collapsing to the smooth level by $m = 8$ — **the committed qualitative branch fired on all three legs** (strict interval scoring: the theory’s residual tail at $m \geq 8$ is refuted) (Figs. 6, 7). Eliminations: not periodicity (harmonic vs incommensurate $\Delta = +0.017$), not antipodality (dead in Wave 3), and provably not RS-visible (Ψ_{RS} flat across comb-vs-sinc while $+0.46$ is measured). **The surviving channel is decision-time multiplicity:** lifted combs give each + pattern 2.1–2.3 viable decision regions versus ≈ 1.2 for smooth kernels — the program’s old “sinusoid 2.3” anomaly generalizes to a law: *multiplicity* $\Leftrightarrow$ *lift*, tying the capacity story to the routing story. Broadening sweep (comb $\rightarrow$ smooth interpolation) remains a costed gap (\$4–5) after a budget stop-spend caused by Vast price-cap non-enforcement (\$11.88 booked vs \$10 cap; incident logged).

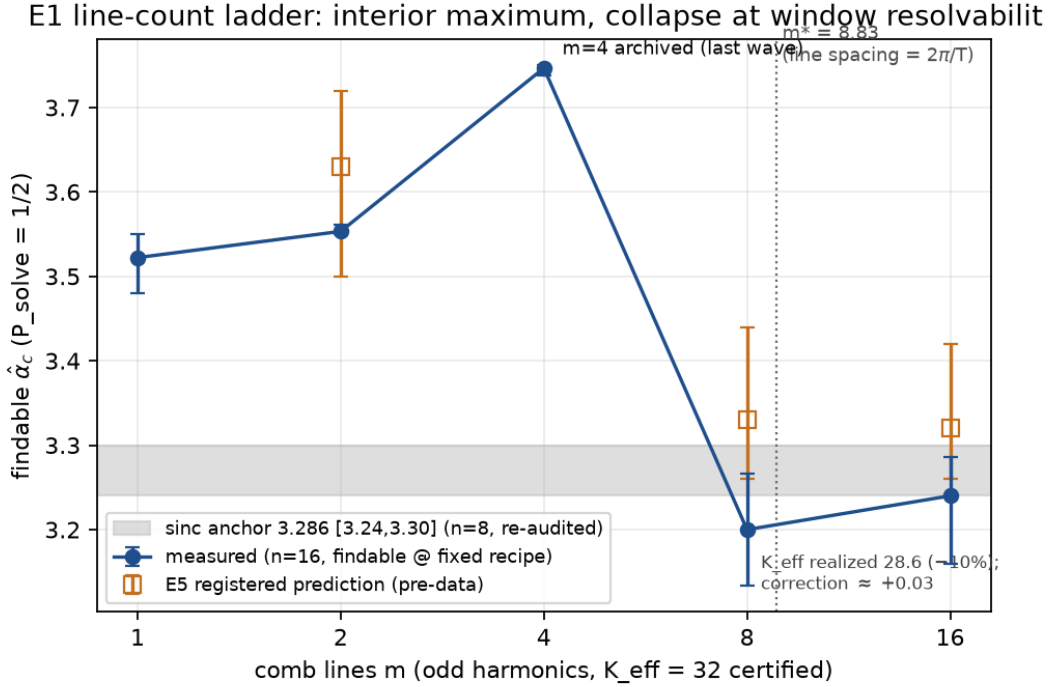


Figure 6: The m -ladder at fixed certified $K_{\text{eff}} = 32$: an interior capacity maximum at $m \approx 3$ –4 lines.

3 Constrained temporal baselines: capacity nulls, learner effects (WS-BASELINE-2)

A learnable baseline $b(t)$ (static input neurons) was studied under a family of constraints (sign, norm budget, balance, sparsity, smoothness, frozen oscillation), theory-first with machine-checked derivations and codex adversarial passes (derivation 16).

Capacity level: one organizing principle. Only the uniform (DC) mode of $b(t)$ has a first-order effect, and *tonic inhibition owns it*: the excitatory-only optimum is exactly $m = 0$ (silent,

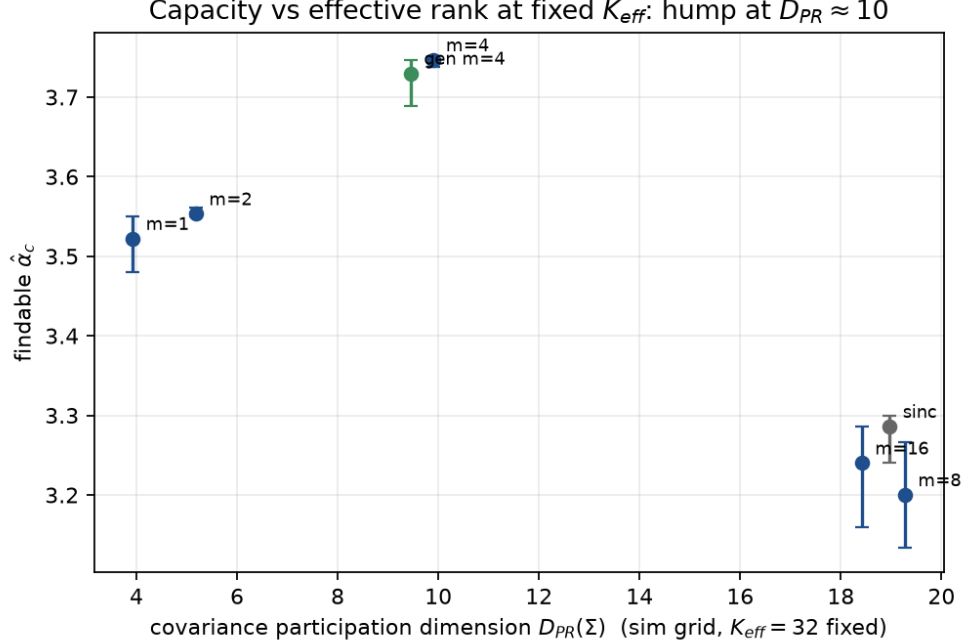


Figure 7: Collapse against effective covariance rank: the lift dies where lines become window-unresolvable, as predicted pre-data.

proven by supermodularity + ray monotonicity), the balance constraint kills the gain identically, norm budgets are monotone with no interior optimum, and sparsity/smoothness reduce to DC overlap. Every constraint is a null or near-null *for capacity*.

Learner level: constraints are live. Three registered outcomes: (i) **Sign reversal confirmed**: frozen random offsets that *cost* -0.19 at the standard operating point *gain* $\geq +0.113$ (point $+0.17$, joint $p = 1.5 \times 10^{-5}$) above the theory’s predicted zero-crossing of the shape-mode curvature — a reversal no other program mechanism predicts (Fig. 8). (ii) The registered “major surprise” branch fired for the balance constraint: GD-*learned* shape offsets produce gains (4/4 vs anchor 0/4 at $\alpha = 4.3$) exactly where both capacity criteria price shape at zero — pending a \$0.9 16-seed confirmation (Fig. 9). (iii) Unconstrained baseline learning is *harmful* (uniform-mode runaway, $m \rightarrow -2.5$): the constraints act as necessary regularization. RS’s silence prediction fails at the GD level (tonic excitation is retained), while the annealed criterion had failed on the tilt signs — neither criterion owns the learner.

Criterion adjudications completed. The tilt anchor is now bracketed (4.15) and the λ -hypothesis refuted on all three of its in-flight tests; the findable surface’s anisotropy against RS is measured directly: the uniform direction responds at $2.7 \times [2.4, 3.2]$ the RS functional, shape directions at $0.49 \times [0.31, 0.65]$ (Fig. 10). The f-sweep span test completed: $\Delta(1) \approx -0.1$ (rigorous $< +0.2$ at 98%) — **both the annealed-cluster span (+0.49) and the RS span (+5.07) are refuted at the findable level** (Fig. 11). **Conclusion: no existing analytic criterion describes the findable surface; it is its own object, now with measured coefficients.** Spend: \$3.79 of \$7.

Predicted sign reversal CONFIRMED: the same frozen offsets cost cap at the anchor but pay at $\tilde{m} = -0.30$ (c_H zero-crossing at $\theta_K + 0.14$)

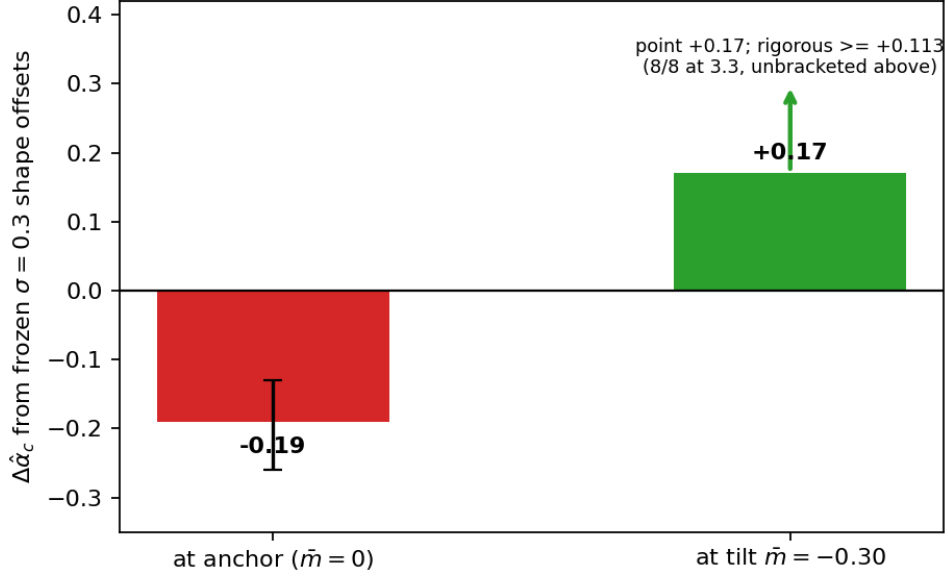


Figure 8: The predicted sign reversal: frozen offsets flip from cost to benefit across the theory’s curvature zero-crossing.

Constrained learnable baselines: capacity theory says all-null;
paired cells: constrained arms never hurt, unconstrained hurts (see gate table)

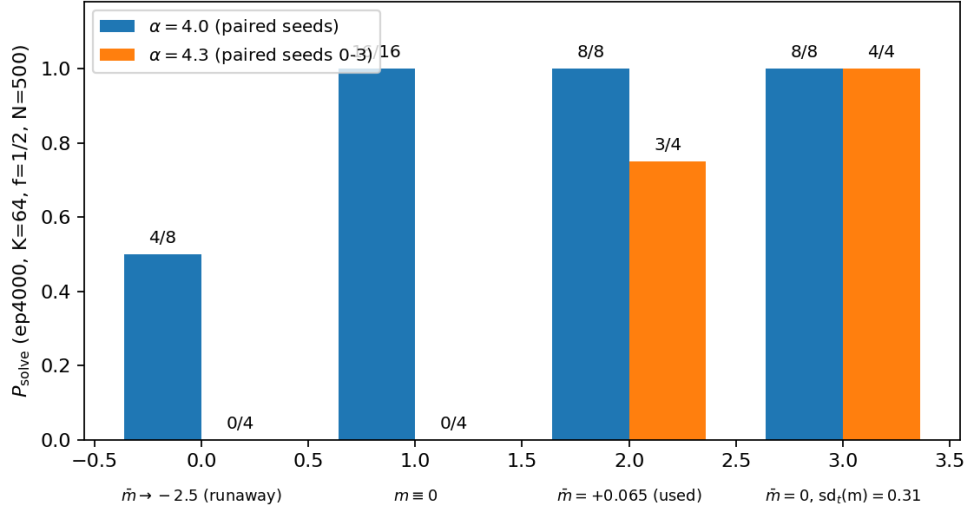


Figure 9: Constrained baseline learning: balance-constrained shape learning helps, unconstrained learning runs away, excitatory-only tracks the anchor.

GD's findable surface is anisotropic vs the RS functional
 arm: $c=0.128$ [0.08,0.17] vs RS 0.264; tilt wings: $2.7\times$ [2.4,3.2])

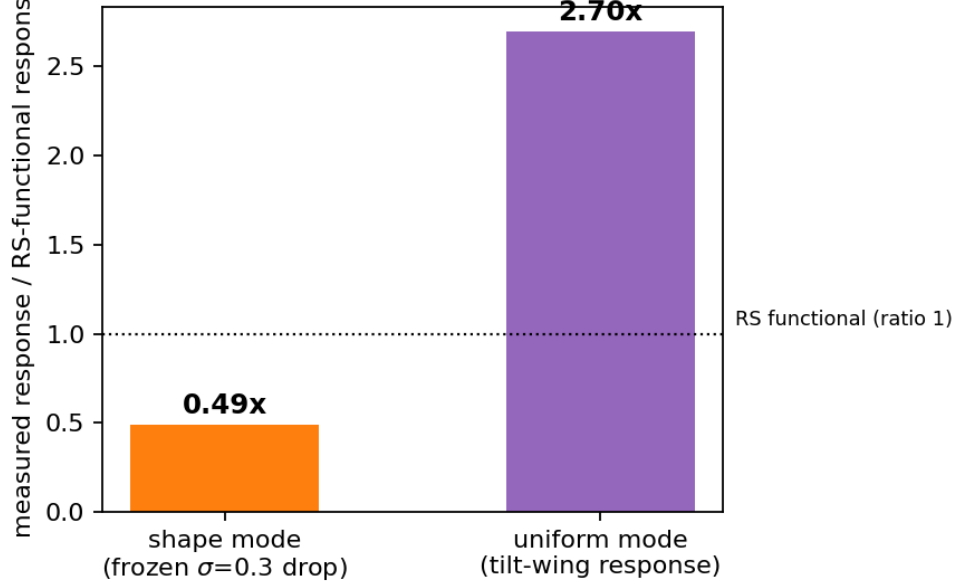


Figure 10: Measured anisotropy of the findable surface relative to the RS functional.

$\Delta(1)$: point estimate -0.1 , rigorous bound $< +0.2$ (98%)
 (cluster $+0.49$ and RS $+5.07$ refuted, findable level)

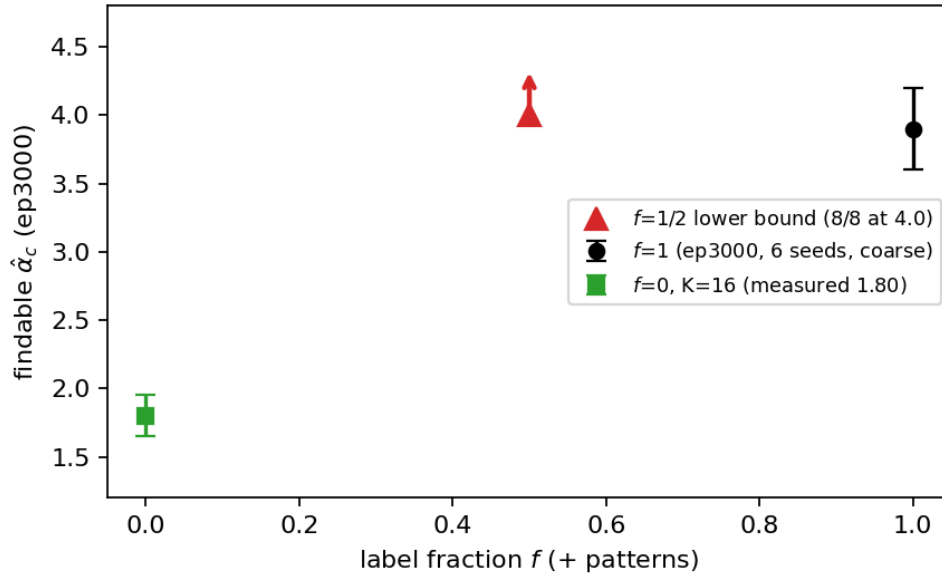


Figure 11: f -sweep completion: both criteria's span predictions are refuted at the findable level.

4 Synthesis

Wave 4 completes an inversion begun in Waves 1–3. The solution space is navigable and its rooms plentiful; the learner even *enters* solution-containing rooms constantly. What fails is a specific, now-named mechanism — ejection by the global update rule before within-room settling — and what no analytic criterion captures is the surface this dynamics actually traces, whose anisotropy is now measured. Meanwhile capacity itself acquired a new organizing variable (effective rank / decision-time multiplicity) and a clean biological statement (tonic inhibition is the only first-order baseline channel). The queued Wave-5 experiment — a within-room feasibility check before exit updates — would convert the autopsy’s diagnosis into a constructive training rule.

Provenance. All claims verifier-gated (independent recomputation); state under `experiments/foundations_{aut}` and `lit/tempotron-reductions/derivations/16-constrained-offsets.md`; pre-registrations and amendments logged before data throughout. Wave-4 lab spend $\approx$ \$16.3; program total $\approx$ \$42 of a \$60 authorization. Known incidents: Vast price-cap non-enforcement (twice; stop-spend honored), an empty GPU market blocking one costed at-scale run, and two detector mis-specifications amended on the record.